4.6 The Fundamental Theorem of Algebra

Name _____ Period ____ Date _____

Algebra II 4.6 · Student sheet

I can…

- use the Fundamental Theorem of Algebra to find the number of solutions of a polynomial equation.
- find all the zeros of a polynomial function and write polynomials from their zeros.
- use Descartes's Rule of Signs.

Vocabulary · fill in the missing word

Fundamental Theorem of Algebra — a _____ equation of degree $n > 0$ has exactly n solutions, counting repeated and imaginary solutions

Complex Conjugates Theorem — if $a + bi$ is a zero of a polynomial with real _____, so is $a - bi$

Descartes's Rule of Signs — the number of _____ real zeros is the number of sign changes in $f(x)$ or less than that by an even number; use $f(-x)$ for negative zeros

Part 1 · Count the zeros

Function	Degree	Real zeros (graph)	Factored form	Imaginary zeros
$x^3 - x$				
$x^3 + x$				
$x^4 - 1$				
$x^4 + 1$				

Part 2 · Notes: fill in the blanks

A polynomial equation of degree n has exactly _____ solutions.

Imaginary zeros come in _____ pairs: $a + bi$ and _____.

Descartes's Rule: count the sign changes in _____ and in _____.

Part 3 · Examples (work them with the class)

Example 1

How many solutions does $x^3 + 64x = 0$ have? Find them.

Example 2

Find all zeros of $f(x) = x^4 + 2x^3 + 4x^2 + 8x$.

Example 3

Write a polynomial function f of least degree with leading coefficient 1 that has the zeros 3 and 2i.

Example 4

Find the possible numbers of positive real zeros, negative real zeros, and imaginary zeros of $f(x) = x^3 - 2x^2 + 3x + 5$.

Part 4 · Partner practice

Partner 1

Write a polynomial function of least degree with leading coefficient 1 that has the zeros 1 (repeated) and 4i.

Partner 2

How many solutions does $x^5 - x = 0$ have? Find them.